

AMENDMENT NO. 2 DECEMBER 1995.

TO

IS 5831 : 1984 SPECIFICATION FOR PVC INSULATION AND SHEATH OF ELECTRIC CABLES

(First Revision)

(Page 4, clause 1.2) -- Add the following matter between 'Type C' and 'Type S11':

'Type D' General purpose insulation for maximum rated conductor temperature 70°C intended for flexible cables with rated voltage U_0/U up to and including 3.3/3.3 kV.'

(Page 4, clause 1.2) -- Add the following new para for Type S13 Compound:

'Type S13 General purpose sheath intended for use in flexible cables operating at maximum rated conductor temperature 70°C.'

(Page 7, Table 1, col 7, Sl No. (xiii)) — Substitute '60' for '(see Note 4)'.

(Page 7, Table 1, Note 4) Delete.

(Pages 5, 6 and 7, Table 1) Add the following test requirements for the new insulation type 'Type D' under column heading 'Type of Insulation' as col 7 of this table and renumber the existing col 7 as col 8 :

Table 1 Test Requirements for PVC Insulation
(Clause 4.1)

Sl No.	Test	Unit	Type of Insulation
			D
(1)	(2)	(3)	(7)
i)	Volume resistivity, $M\Omega$		
	a) at 27°C (see Note 1)	Ohm-cm	1×10^{10}
	b) at maximum rated temperature (see 1.2)	Ohm-cm	1×10^9
ii)	Insulation resistance constant, $M\Omega$		
	a) at 27°C (see Note 1)	$M \Omega$ km	3.67
	b) at maximum rated temperature (see 1.2)	$M \Omega$ km	0.004
iii)	Without ageing		
	a) tensile strength, MN	N/mm ²	10
	b) Elongation at break, $M\%$	Percent	150

Price Group 1

Table 1 (Continued)

Sl No.	Test	Unit	Type of Insulation
(1)	(2)	(3)	(4)
iv)	After ageing in air oven		
a)	Treatment { Temperature (tolerance $\pm 2^{\circ}\text{C}$) Duration	$^{\circ}\text{C}$ days	70 7
b)	Tensile strength		
1)	Value after ageing, <i>Min</i>	N/mm ²	10
2)	Variation, <i>Max</i>	percent	± 20
c)	Elongation at break		
1)	Value after ageing, <i>Min</i>	percent	150
2)	Variation, <i>Max</i>	percent	± 20
v)	Loss of mass in air oven		
a)	Treatment { Temperature (tolerance $\pm 2^{\circ}\text{C}$) Duration	$^{\circ}\text{C}$ days	80 7
b)	Loss of mass, <i>Max</i>	mg/cm ²	2
vi)	Hot deformation test		
a)	Test temperature (tolerance $\pm 2^{\circ}\text{C}$)	$^{\circ}\text{C}$	30
b)	Time under load (see Note 2)		
1)	First case	hours	4
2)	Second case	hours	
c)	Depth of Indentation, <i>Max</i>	percent	65
vii)	Heat shock test		
a)	Treatment { Temperature (tolerance $\pm 2^{\circ}\text{C}$) Duration	$^{\circ}\text{C}$ hour	150 1
b)	Visual examination	---	No signs of cracks or scales
viii)	Shrinkage test		
a)	Treatment { Temperature (tolerance $\pm 2^{\circ}\text{C}$) Duration	$^{\circ}\text{C}$ minutes	150 15
b)	Shrinkage, <i>Max</i>	percent	6
ix)	Cold bend test (for diameter $\leq 12.5 \text{ mm}$)		
a)	Test temperature (tolerance $\pm 2^{\circ}\text{C}$)	$^{\circ}\text{C}$	- 15
b)	Visual examination	—	No signs of cracks or scales
x)	Cold impact test (for diameter $> 12.5 \text{ mm}$)		
a)	Test temperature (tolerance $\pm 2^{\circ}\text{C}$)	$^{\circ}\text{C}$	- 5
b)	Visual examination	—	No signs of cracks or scales

Table 1 (Concluded)

Sl No.	Test	Unit	Type of Insulation D (7)
(1)	(2)	(3)	(7)
xi)	Colour fastness to daylight exposure, <i>Min</i>	rating	4
xii)	Colour fastness to water		
	a) Treatment { Temperature (tolerance $\pm 2^{\circ}\text{C}$) Duration	$^{\circ}\text{C}$ hours	70 48
	b) Visual examination	—	Free from traces of colour
xiii)	Thermal stability		
	a) Test temperature (tolerance $\pm 0.5^{\circ}\text{C}$)	$^{\circ}\text{C}$	200
	b) Time, <i>Min</i>	Minute	80
xiv)	Bleeding and blooming test		
	a) Treatment { Temperature (tolerance $\pm 2^{\circ}\text{C}$) Duration	$^{\circ}\text{C}$ hours	50 72
	b) Observation	--	No appreciable staining of indicator compound or filter paper
xv)	Water absorption test (gravimetric)		
	a) Treatment { Temperature (tolerance $\pm 2^{\circ}\text{C}$) Duration	$^{\circ}\text{C}$ days	— —
	b) Water absorbed, <i>Max</i>	mg./cm ²	—

[Page 10, Table 2, col 6, Sl No. (xi)] --- Substitute '60' for '(See Note 3)'.

(Page 10, Note 3) - Delete.

(Pages 8, 9 and 10, Table 2) -- Add the following Test Requirements for the new PVC Compound 'Type ST3' under column heading 'Type of Sheath' as col 6 of this table and renumber the existing col 6 as col 7 :

**Table 2 Test Requirements for PVC Sheath
(Clause 4.1)**

Sl No.	Test	Unit	Type of Sheath ST3 (6)
(1)	(2)	(3)	(6)
i)	Without ageing		
	a) Tensile strength, <i>Min</i>	N/mm ²	10.0
	b) Elongation at break, <i>Min</i>	percent	150

Table 4 (Continued)

Sl No. (1)	Test (2)	Unit (3)	Type o (4)	Health (5)
ii)	After ageing in air oven			
	{ Temperature	°C	7	80
	a) Treatment { (tolerance $\pm 2^{\circ}\text{C}$)			
	{ Duration	days		7
	b) Tensile strength			
	1) Value after ageing, M_{fin}	N/mm ²		100
	2) Variation, M_{var}	percent		± 20
	c) Elongation at break			
	1) Value after ageing, M_{fin}	percent		150
	2) Variation, M_{var}	percent		± 20
iii)	Loss of mass in air oven			
	{ Temperature	°C		80
	a) Treatment { (tolerance $\pm 2^{\circ}\text{C}$)			
	{ Duration	days		7
	b) Loss of mass, M_{loss}	mg/cm ²		20
iv)	Hot deformation test			
	a) Test temperature	°C		70
	(tolerance $\pm 2^{\circ}\text{C}$)			
	b) Time under load (see Note 1)			
	1) First case	hours		1
	2) Second case	hours		6
	c) Depth of indentation, M_{max}	percent		65
v)	Heat shock test			
	{ Temperature	°C		150
	a) Treatment { (tolerance $\pm 2^{\circ}\text{C}$)			
	{ Duration	hour		1
	b) Visual examination	—		No signs of cracks or scales
vi)	Shrinkage test			
	{ Temperature	°C		150
	a) Treatment { (tolerance $\pm 2^{\circ}\text{C}$)			
	{ Duration	minutes		15
	b) Shrinkage, M_{sh}	percent		6
vii)	Cold bend test for diameter ≤ 12.5 mm			
	a) Test temperature	°C		— 15
	(tolerance $\pm 2^{\circ}\text{C}$)			
	b) Visual examination	—		No signs of cracks or scales
viii)	Cold impact test for diameter >12.5 mm			
	a) Test temperature	°C		— 5
	(tolerance $\pm 2^{\circ}\text{C}$)			
	b) Visual examination			No signs of cracks or scales

Table 2 (Concluded)

Sl No. (1)	Test (2)	Unit (3)	Type of Sheath (6)
ix)	Colour fastness to daylight exposure, <i>Mfn</i>	rating	4
x)	Colour fastness to water		
	a) Treatment { Temperature (tolerance $\pm 2^{\circ}\text{C}$)	$^{\circ}\text{C}$	70
	{ Duration	hours	48
	b) Visual examination	—	Free from traces of colours
xi)	Thermal stability		
	a) Test temperature (tolerance $\pm 0.5^{\circ}\text{C}$)	$^{\circ}\text{C}$	200
	b) Time, <i>Mfn</i>	Minutes	40
xii)	Bleeding and blooming		
	a) Treatment { Temperature (tolerance $\pm 2^{\circ}\text{C}$)	$^{\circ}\text{C}$	50
	{ Duration	hours	72
	b) Observation		No appreciable stain- ing of indicator com- pound or filter paper

(ETD 09)

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